
ABILAKSHAN VADIVEL

SURVEYOR | GIS ANALYST

Professional Interests: Surveying | GIS Analysis | Drone Survey
| Web GIS | GeoAI

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Jaffna, Sri Lanka

*B.Sc. (Hons) in Surveying Sciences,
specializing in GIS*

PROFESSIONAL SUMMARY

Motivated Surveying Sciences graduate with GIS specialization and practical experience in land demarcation, GNSS-based field support, spatial data preparation, and survey-related mapping workflows. Experienced in CAD-to-GIS conversion, land information system development, thematic map production, and geospatial database preparation using both Esri and open-source GIS tools. Bringing a combined Surveyor and GIS Analyst profile, with the ability to support both field surveying operations and office-based GIS, mapping, and data management tasks.

TECHNICAL PROFILE

Surveying & GIS Skills:

Boundary Surveying | Topographic Surveying | Engineering Surveying | GNSS-based Field Data Collection | CAD-to-GIS Conversion | Spatial Data Cleaning & Digitizing | Spatial Analysis | Thematic Mapping | Land Information Systems | Web GIS Layer Preparation | WMS/WFS Publishing

Software:

ArcGIS Pro | ArcGIS Online | ArcGIS Experience Builder | QGIS | PostgreSQL | PostGIS | GeoServer | Leaflet | React | MapLibre | AutoCAD | Civil 3D | CloudCompare | Pix4D Mapper | Google Earth Engine | VS Code | MS Office

Instrument Exposure:

GNSS/GPS Receivers | Total Station | Auto Level | Theodolite | Drone Platforms | Terrestrial Laser Scanner (TLS) | Ground Penetrating Radar (GPR)

WORK EXPERIENCE

Planning Intern (Surveyor & GIS Analyst) | Riyan Private Limited

Dec 2025 - March 2026

| Maldives

Land Information System (LIS) using Esri Products (ArcGIS Pro, ArcGIS Online)

- Prepared and organized project geospatial data, including shapefiles, CAD-based inputs, and attribute information, for LIS development and land-use mapping workflows.
- Converted CAD files into GIS-ready layers and created and edited point, polyline, and polygon features under required land-use categories in ArcGIS Pro.
- Developed web-based LIS views by publishing datasets through ArcGIS Online and configuring Web Maps, Web Scenes, and Experience Builder components for interactive access.

- Supported layer organization, symbology, pop-up configuration, and 3D map preparation to present land-use and planning information more effectively.

Land Information System (LIS) using Open-Source GIS & Spatial Database (QGIS, PostgreSQL/PostGIS, GeoServer)

- Maintained consistent data structure, layer organization, and project outputs while transitioning LIS workflows from Esri platforms to open-source tools.
- Organized project files and QGIS layers, connected QGIS with PostgreSQL through pgAdmin, enabled PostGIS, and imported prepared layers into the spatial database.
- Structured portal-ready datasets and supported GeoServer publication workflows by preparing layers and creating basic WMS and WFS services for required data.
- Assisted Web GIS portal development using Leaflet, HTML, CSS, and JavaScript, including layer controls, legend visibility, and measurement tool integration.

Land Demarcation & GNSS Field Support

- Reviewed approved plans, carried out site reconnaissance, and checked demarcation areas before field implementation for residential, tourism, and agriculture zones.
- Assisted GNSS-based field setup, observations, boundary reference checking, and block demarcation activities during on-site land demarcation work.
- Maintained field notes, daily progress records, and supporting documentation to ensure smooth continuation of field tasks and office reporting.
- Supported field completion reporting using reference photographs and drone imagery to communicate site status and completed works clearly.

Coastal / Sediment Analysis

- Supported coastal sediment laboratory work for beach nourishment and coastal protection studies by identifying, labeling, and organizing sand samples.
- Carried out sample washing, drying, weighing, and initial preparation tasks while maintaining sample tracking records and laboratory documentation.
- Assisted sieve analysis using multiple sieve sizes and recorded retained material values for grain-size distribution assessment.
- Contributed to documentation used to evaluate suitable sand material for coastal protection and construction-related applications.

GIS Outputs, Heat Maps, Harbour Digitization & Drone Exposure

- Produced and revised Resilient Schools heat maps by rechecking mapped point alignment, correcting layout issues, and updating outputs based on in-charge feedback.
- Prepared school sketches, open area versus built-up support layers, and time-based heat map outputs for multiple schools and reporting needs.

- Supported harbor digitization and GIS feature editing by creating, updating, and structuring spatial layers for planning and presentation purposes.
- Gained exposure to Pix4D Mapper workflows and reviewed orthomosaic, point cloud, DTM, and DSM outputs to understand drone image processing deliverables.

GeoAI for Disaster Resilience - Undergraduate Fellowship | ADRIMP

Feb 2026 - Present

- Applied GeoAI, spatial analytics, and risk-modeling concepts to support disaster resilience and risk-reduction planning.
- Worked with geospatial datasets, hazard mapping approaches, and decision-support frameworks for emergency preparedness.
- Strengthened knowledge in resilience-oriented geospatial analysis and GeoAI-based spatial interpretation.

Management Assistant (IT) - Internship | Institute of Medical Sciences, Manipal

Mar 2021 - Aug 2021

- Provided IT support and assisted with database-related tasks to help maintain smooth organizational operations.

SURVEYING PROJECTS

Topographic Survey & Control Establishment | North Axis (Pvt) Ltd

- Participated in a professional topographic survey for a water tank and sump project in Meesalai, Jaffna.
- Established three primary control points and collected field data for contour plan preparation.

Engineering Survey for Reservoir Construction | Academic Field Project

- Conducted engineering and topographic survey tasks for a proposed reservoir and supported access-road design using field survey data.

Survey Camp | Karagasthalawa Maha Vidyalaya, Belihuloya

- Completed topographic and boundary surveys, established control points using static GPS, and prepared site plans for future development.

Advanced Surveying Technology Workshops

- Gained exposure to drones (LiDAR and multispectral), Terrestrial Laser Scanning (TLS), and Ground Penetrating Radar (GPR) through technical workshop sessions.

GIS PROJECTS

Land Information System / Interactive Web GIS Project

- Prepared and organized shapefiles, satellite imagery, drone imagery, and CAD data within a common GIS workflow.
- Contributed to layer symbology, labels, attribute tables, pop-ups, web-layer publishing, and interactive map configuration.

- Supported LIS implementation using ArcGIS-based workflows and open-source tools including QGIS, PostgreSQL/PostGIS, GeoServer, Leaflet, React, and MapLibre.

Landslide Vulnerability Mapping

- Produced a landslide vulnerability map for Matale District using GIS-based spatial analysis and the Analytical Hierarchy Process (AHP).

DEM Comparison and Geostatistical Analysis

- Generated and compared DEMs derived from digitized contours with SRTM and ASTER data and completed geostatistical analysis exercises in R.

Final Year Undergraduate Research

- Completed final-year undergraduate research on GeoAI-based spatial feature detection using geotagged data and multi-temporal Sentinel-2 imagery.
- Applied DeepLabV3+ modelling and achieved 73% accuracy in the completed final model results.

Additional GIS Work

- Created an interactive StoryMap to visualize the impacts of the 2019-2020 Australian bushfires.
- Developed a dynamic dashboard for real-time weather visualization and monitoring.

EDUCATION

B.Sc. (Hons) in Surveying Sciences, specializing in GIS | Sabaragamuwa University of Sri Lanka

- Completed academic requirements | GPA: 3.4 / 4.00 (up to 3rd Year, 2nd Semester) | Faculty of Geomatics

Full Stack Web Development | Uki Coding School

Jun 2020 - Nov 2020

- Online Cohort

- Covered ReactJS, NodeJS, and MongoDB fundamentals.

G.C.E. Advanced Level - Physical Science | St.

2019

John's College, Jaffna

- Results: 2B, 1C

PROFESSIONAL MEMBERSHIPS

- Royal Institution of Chartered Surveyors (RICS) - Professional Number(Student): 0994438
- Geo Mappers
- ISPRS Student Consortium

CERTIFICATES

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| • Spatial Data Science: The New Frontier in Analytics - MOOC | • Creating an ArcGIS StoryMaps Briefing |
| • Make an Impact with Modern Geo Apps - MOOC | • Introduction to Regression Analysis Using ArcGIS Pro |
| • Going Places with Spatial Analysis - MOOC | • Building a Regression Model Using ArcGIS Pro |
| • GIS for Climate Action - MOOC | • Python for Beginners |

- Cartography - MOOC
- Telling Stories with GIS Maps

- Foundations of Project Management

EXTRACURRICULAR ACTIVITIES & LEADERSHIP

- Inter-University Badminton Team Vice-Captain (2025), Member (2022-2025)
- Inter-University Table Tennis Runners-Up (2022), Member (2022-2025)
- School Badminton Team Captain (2017), Member (2012-2018)
- School Volleyball Team Captain (2018), Member (2017-2018)
- School Cricket Team Member (2011-2018)
- School Prefect (2016-2018)
- Best Cricket Team in Northern Province (2017)
- All-Island School Cricket Under-13 Runners-Up (2013)

REFERENCES

Prof. RMKGSPB Koswatte

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